Food Expenditures Consulting

Problem Background

For this project you are hired as a consultant to the National Restaurant Association to investigate how customers eating out behavior is influenced by their income levels. If restaurant food is a “normal” good, one would expect consumption (as measured by spending) to increase as income increases. This relationship has implications for projections of food demand and restaurant ownership as a viable career as incomes grow or as countries develop and become wealthier. However, just because people will spend more at restaurants as their income increases, on average, the relationship is likely more complicated due to high degree of variability for high income individuals. Also of interest is how demand for “quality” and “convenience” changes as income changes, which relates to whether people spend more money on food at home vs. away from home as income increases.

To accomplish this consulting task, you gathered the dataset *FoodExpenses.txt* from the Food Demand Survey (FooDS) on $n = 523$ households income and associated expenses on “eating out”. Specifically, *FoodExpenses.txt* contains the following variables:

Variable Name	Description
Income	Annual household income (in thousands)
EatingOut	Average weekly expenditure on food not cooked at home.

For this project, you are to write a technical report to the financial advisers to the National Restaurant Association. These advisers are finance undergrads so they know standard regression but that is the extent of their modeling experience. Your report should cover the following topics:

1. Introduction
    - Provide a basic recap of why you were hired and the goals of your consulting project
2. Main Conclusions
    - Describe the primary conclusions about the relationship between income and food expenditures. What relationships did you find?
    - The advisers specifically hypothesize that a “healthy” restaurant economy should see increases of about $0.50 or more per week for each $1000 increase in income. Based on the data you have, do you think this is a healthy economy?
    - Advisers are expecting incomes to generally shift upward in time (income generally increases by 3-4% each year). Make sure you provide some predictions on what an upward shift in income over the next 5 years or so could do to food expenditures (including the variance not just the amount spent on EatingOut).
3. Modeling Details
    - Justify your hiring by showing that standard regression models won’t work for this situation so they needed to hire you and should hire you again.
    - Describe this new “advanced” model you are using in mathematical detail and how you used it to arrive at the conclusions you came to in the following section. In doing so, these advisers

don’t know linear algebra so try to avoid that if possible.
  - Justify (i.e validate) that this model is a trustworthy model for the answers you wrote in the previous section.

Suggestions on Use of AI

If you want to use AI to help you (which I recommend), here are some useful strategies that will still make sure you are learning and that your final work reflects *your* analysis. First, upload this assignment and your dataset (and, if relevant, your current draft or rendered output) so AI has the full context.

1. **Use AI to plan exploration and figures**
   - *Prompt idea:* “Based on the goals of this project, what exploratory summaries and diagnostic plots would be most appropriate? What patterns or red flags should I watch for as I explore the data/model?”
2. **Use AI as a conceptual tutor**
   - *Prompt idea:* “I am not sure how to answer the question on seeing more than 0.50 cents per $1000 increase. Can you help me understand how to approach this?”
3. **Use AI for coding help, but require explanation**. To give AI some course context, it can also be helpful to upload your HW code or my skeleton code from my lecture notes so it can use functions you are familiar with.
   - *Prompt idea:* “I want to fit (insert type of model you are thinking here). Help me generate R code to do this, but explain what each piece does and why it’s needed. Use the skeleton code from the instructor as much as possible. Also tell me how I should interpret the output.”
4. **Use AI as an editor for your writing (not a results-generator)**
   - *Prompt idea:* “Help me organize, expand, and polish this section **based only on the results shown in my output**. Do not invent numbers, conclusions, or claims—only use what I have already computed.”
5. **Use AI to find gaps or missed opportunities**
   - *Prompt idea:* “Review my draft and list things that would strengthen the analysis or make the report clearer. Please give suggestions, but do not invent results I haven’t computed.”

Important guardrails:

- Think of AI as a group member that an contribute to the overall project but you still need to check its work.
- Keep AI in a supporting role: planning, tutoring, code explanation, and editing.
- Always verify code and interpretations yourself.
- Don’t include AI-generated numbers/claims unless you can point to your own computed output that supports them. If you don’t know where AI got a result, ask it to explain or show the calculation.
- Save a short record of prompts you used (helps you stay consistent and transparent).